SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING
Program Name: B. Tech	**Assignment Type: Lab**	**Academic Year:**2025-2026

Q.No.	Question	*Expected Time to complete*
1	**Lab 21 – Collaborative AI Coding: Pair Programming and Version Control Integration** **Lab Objectives:** • Learn to use AI as a **pair programmer** for writing, reviewing, and optimizing code. • Understand how to integrate AI-generated code into **version control systems (Git)**. • Practice collaborative workflows such as **commit messages, branching, and merging**. --- **Task 1 – AI-Assisted Pair Programming** • Use AI to generate an initial program (e.g., a simple To-Do List Manager). • Ask AI to suggest **enhancements** (such as error handling, additional features, or refactoring). • Compare the original and improved versions to understand the role of AI in collaboration. --- **Task 2 – Version Control Integration** • Initialize a Git repository for your project. • Perform the following steps: 1. Create and switch to a new branch. 2. Add AI-assisted modifications. 3. Commit with a meaningful message. 4. Merge changes back into the main branch. • Reflect on how version control supports collaborative development with AI assistance. --- **Task 3 – AI for Commit Message Generation** • Use AI to generate **concise, meaningful commit messages** based on the changes made.	**Week11 – Monday**

	• Compare AI-generated messages with human-written ones. • Discuss how AI can improve **documentation and traceability** in Git history.	
	Task 4 – Pair Programming Simulation • Student A writes the initial version of a program. • Student B, with AI assistance, suggests **improvements or adds features**. • Both students collaborate using **Git branching and merging** to integrate their work.	
	Task 5 – AI-Assisted Debugging • Create a program with logical or syntax errors. • Use AI to identify and fix the errors. • Compare the buggy and corrected versions.	
	Task 6 – Feature Expansion using AI • Develop a basic application (e.g., Calculator). • Use AI to add advanced features (history, validation). • Analyze improvements.	